

DepthWizard — Final Four-Landscape Benchmark Report

Single-view height estimation and 3D terrain reconstruction

Benchmark completion

All four requested landscape categories were completed: Urban, Sparse, Hilly and Forested. Each scene was processed through the DepthWizard metric workflow and reference validation.

Scene	Landscape	RMSE (m)	MAE (m)	R ²	Pearson R	Bias (m)	Valid (%)
01	Urban	35.306	27.494	0.014	0.120	0.000	100.0%
02	Sparse	5.850	4.799	0.034	0.183	0.000	33.8%
03	Hilly	69.473	59.288	0.117	0.342	0.000	100.0%
04	Forested	13.515	10.987	0.356	0.596	0.000	21.4%

Key observations

- Sparse produced the lowest RMSE (5.850 m) but only 33.8% valid coverage.
- Forested produced the strongest correlation in this set (Pearson R = 0.596) and R² = 0.356, but valid coverage was only 21.4%.
- Hilly produced the highest RMSE (69.473 m), indicating substantially greater reconstruction difficulty for this scene.
- Urban achieved 100% valid coverage, but the RMSE was 35.306 m and correlation was low (Pearson R = 0.120).

SIH-safe interpretation

Recommended wording: “DepthWizard successfully completed an end-to-end four-landscape reference-validation workflow covering urban, sparse, hilly and forested scenes. The results demonstrate the complete processing, metric calibration, reference validation, spatial error mapping and export pipeline. Measured accuracy varies by landscape and valid reference overlap.”

Important qualification: The references used are USGS 1 m elevation DEM products. Because such DEMs are generally bare-earth references while RGB imagery may contain vegetation and buildings, these results should be presented as reference-DEM validation rather than definitive DSM-vs-DSM ground-truth accuracy.

Coverage warning: Sparse (33.8%) and Forested (21.4%) have limited valid overlap. Do not present their accuracy values without showing the valid-coverage percentage.

DepthWizard requirement evidence

Requirement	Evidence from completed workflow
PNG/JPG/GeoTIFF input	Implemented input pipeline; GeoTIFF benchmark scenes processed.
Relative height / monocular depth	Pretrained Depth Anything V2 workflow used.
Metric reconstruction	DEM/GCP calibration and metric DSM generation completed.
3D visualization	Interactive terrain mesh, texture projection and inspection available.
Validation	RMSE, MAE, R ² , Pearson R, Bias, valid coverage and spatial error map.
Exports	Metric DSM GeoTIFF, depth preview, GLB terrain mesh, error map, JSON and HTML report.
Four-landscape benchmark	Urban, Sparse, Hilly and Forested completed.